

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data=pd.read_csv('/content/Gender_Inequality_Index.csv')

numerical_columns = data.select_dtypes(include=['int64',
'float64']).columns
print("Numerical Columns:", numerical_columns)

Numerical Columns: Index(['GII', 'Rank', 'Maternal_mortality',
'Adolescent_birth_rate',
'Seats_parliament', 'F_secondary_educ', 'M_secondary_educ',
'F_Labour_force', 'M_Labour_force'],
dtype='object')

mean=data['GII'].mean()
print(mean)

0.3443764705882353

median=data['GII'].median()
print(median)

0.363

var=data['GII'].var()
print(var)

0.038850496484510964

std_dev=data['GII'].std()
print(std_dev)

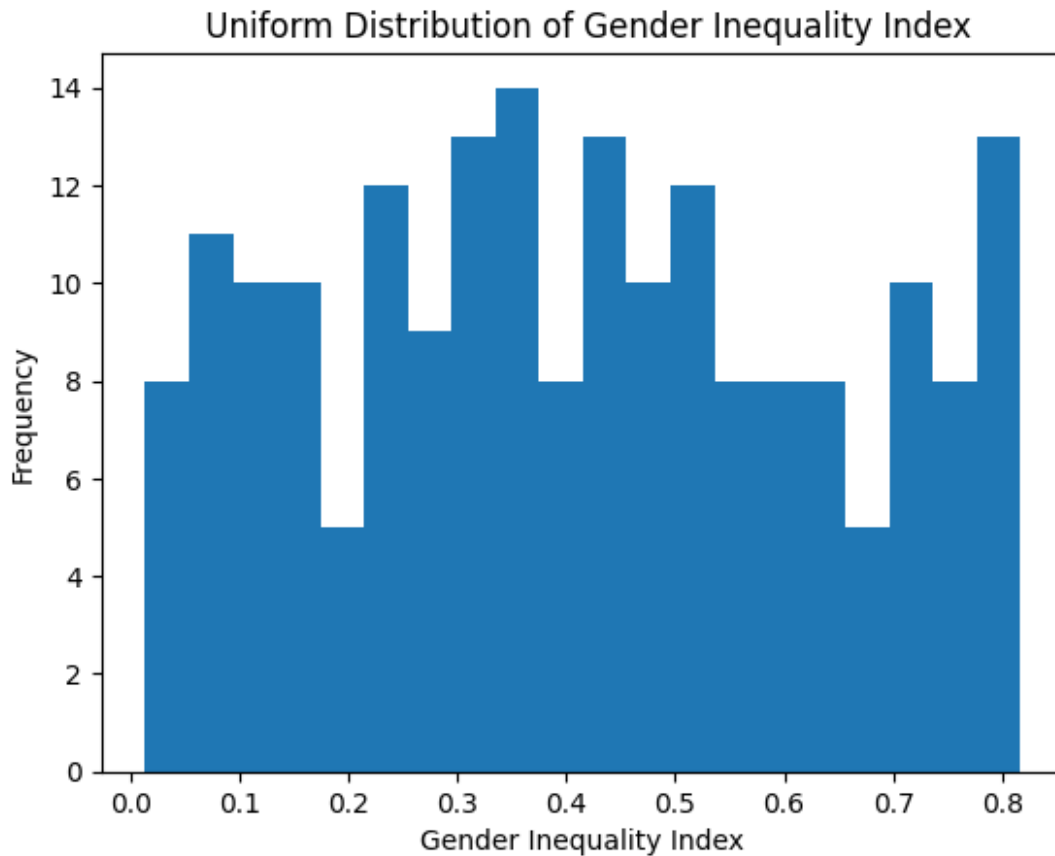
0.19710529288811846

plt.hist(data['GII'],bins=20)
plt.xlabel("Gender Inequality Index")
plt.ylabel("Frequency")
plt.title("Distribution of Gender Inequality Index")
plt.show()

```



```
uniform_gii = np.random.uniform(  
    low=data['GII'].min(),  
    high=data['GII'].max(),  
    size=len(data)  
)  
  
plt.hist(uniform_gii, bins=20)  
plt.xlabel("Gender Inequality Index")  
plt.ylabel("Frequency")  
plt.title(" Uniform Distribution of Gender Inequality Index")  
plt.show()
```



```
corr_flf=data['GII'].corr(data['F_Labour_force'])
corr_mm=data['GII'].corr(data['Maternal_mortality'])
print(corr_flf)
print(corr_mm)

-0.07096963172782948
0.7135147204902433

plt.scatter(data['F_Labour_force'], data['GII'])
plt.xlabel("Female Labour Force Participation Rate")
plt.ylabel("GII")
plt.title("GII vs Female Labour Force Participation")
plt.show()
plt.scatter(data['Maternal_mortality'], data['GII'])
plt.xlabel("Maternal Mortality Ratio")
plt.ylabel("GII")
plt.title("GII vs Maternal Mortality Ratio")
plt.show()
```

